

Midlands Fluid Mechanics Meeting 2026

Scarman Conference Centre, University of Warwick, Coventry, UK

June 15, 2026

It is our pleasure to welcome you to the University of Warwick for the fourth edition of the Midlands Fluid Mechanics Meeting (MFMM 2026). We are pleased to continue this new tradition, following previous editions held at Aston University, Leicester University, and the University of Birmingham from 2023–2025, respectively. MFMM aims to bring together academics from across the Midlands who are engaged in research related to fluid mechanics. The region hosts a substantial body of pio-



neering work in this field, and the meeting is intended to serve as a platform for researchers to share insights from their respective institutions. Equally, it will provide an informal yet productive setting to facilitate the development of new academic collaborations and strengthen regional research networks.

This will be a one-day conference and networking event among the fluid mechanics colleagues from across the Midlands and beyond. We are pleased to say that this edition is the most well-attended MFMM meeting yet – the meeting's growth over the past four years shows no sign of abating, which is testament to the developing strength & depth of fluids research within the Midlands. Therefore, it is not surprising that the meeting was heavily oversubscribed for oral presentations; we would like to sincerely thank those who were willing to give up their talks to ensure that the widest possible range of presentations, including from early career researchers, could be included in the programme.

We thank Warwick's [Digital, data science & AI spotlight](#) and the [School of Engineering](#) for the funding that has enabled this event to take place. The event is being organised under the auspices of Fluid Dynamics Research Centre, a cross-departmental unifying centre for fluid dynamics, and the School of Engineering's Fluids & Thermal Research Cluster

Welcome to Warwick!

With very best wishes,
MFMM 2026 Local Organising Committee

Local Organising Committee

Yongmann Chung

Associate Professor, School of Engineering, University of Warwick

Tom Sykes

Assistant Professor, School of Engineering, University of Warwick

Peter Thomas

Professor, School of Engineering, University of Warwick

Michael Auinger

Associate Professor, WMG, University of Warwick

Duncan Lockerby

Professor, School of Engineering, University of Warwick

Petr Denissenko

Associate Professor, School of Engineering, University of Warwick

Hossam Zeitoun

Reader, WBS, University of Warwick

Feng Mao

Associate Professor, School for Cross-Faculty Studies, University of Warwick

Registration Details

Total Registrations: 74.

Total number of oral presentations: 24 (many more were offered).

Total number of posters: 11.

Conference Venue

The meeting will be held at **Scarman Conference Centre** (What3Words: [///them.punch.oath](https://www.what3words.com/en/@@@them.punch.oath)) on the [main campus](#) of the University of Warwick (CV4 7AL). On Google Maps, it appears as “[Scarman — Warwick Conferences](#)”. Talks will be held in **Space 41** (upstairs), breaks in the “Lounge” accessed via the corridor on the right of reception, and lunch in the restaurant next to the lounge. Tea, coffee, and snacks are available all day in the lounge (including in the morning, so feel free to get a drink before coming to space 41).

While Canley and Tile Hill are the [closest train stations](#), both are small with limited service and are a roughly 40 min walk to Scarman (or require a bus). Visitors to campus are usually better off taking a train to the nearest main-line station, which is **Coventry train station**, and a bus from there. From the South, **Leamington Spa train station** is also an option, but the bus ride is longer than from Coventry. Please be aware that, despite the name, the University of Warwick is on the outskirts of Coventry; Warwick rail station is > 8 miles from campus.

The only last leg public transport option is a local bus. The main University bus stop is called “Interchange”; it is a 5–10 min walk to Scarman, as indicated in the map overleaf. This stop is served by [very regular buses 11, 12X, 14/14A](#) from Coventry train station (5 min walk from the station to the Rail Station Bridge bus stop) or [U1, U1A, 11](#) from Leamington Spa (5 min walk from the train station to the Parish Church bus stop). The geographically closest stop to Scarman is [Scarman Road](#), which is served by a subset of the buses listed above.

For our event, **car parking is free, but only when you book [on this webpage](#) and use the promo code emailed to you** (the code will reduce the usual £8.50 charge to £0.00). Fill in your personal details (including vehicle registration) and check the “Booking Summary” very carefully – especially the date and entry/exit times. Once booked, you must park only in the conference centre car park (P26), which is located behind Scarman at [///merit.liver.co.uk](https://www.merit.liver.co.uk) (What3Words). Your car park booking will be checked with ANPR (no need to display a ticket etc.) and there are no barriers. Parking in the Radcliffe conference centre car park is also fine, in case staff direct you there on the day. Your parking registration does not apply to any other campus car park, which all have ANPR! More information about parking (including EV charging facilities) is available on the [Warwick Conferences website](#).

Dinner

After the conference, we will go to dinner at [Blue Orchid](#) (CV1 3GR), which is a short drive from Scarman Conference Centre (10 minutes or so) towards Coventry. **Drivers who are willing to give lifts to those coming by public transport, please let us know by email how many seats you can spare.** At the restaurant, there is a car park with free parking (requiring registration once we get there).

For those coming by public transport, we will match you with another participant who is able to give you a lift (or we will arrange an alternative provision) on the day. The restaurant is within a 10 min walk of Coventry train station for your journey home.

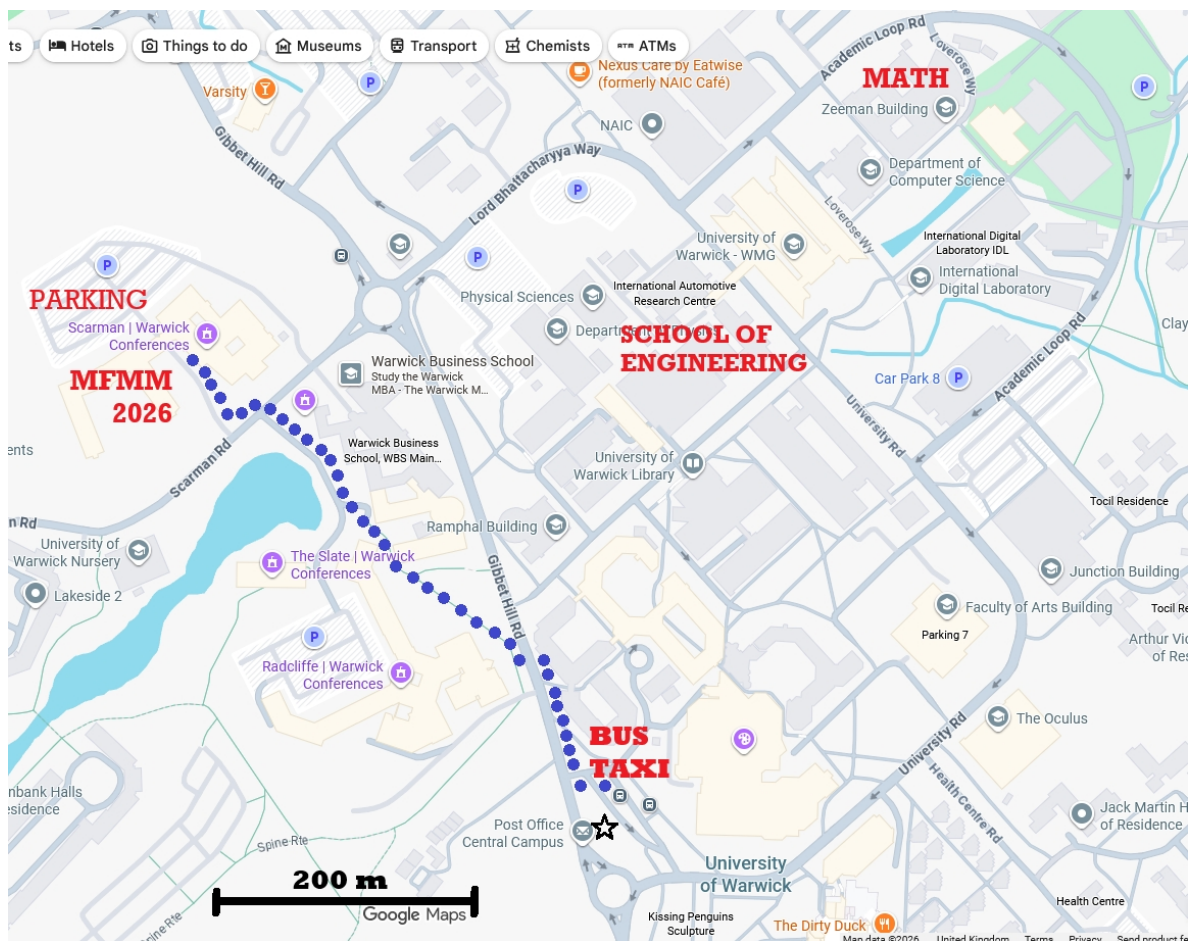


Figure 1: The meeting will be held at **Scarman Conference Centre** (What3Words: [///them.punch.oath](https://www.what3words.com/?w3w=them.punch.oath)) on the **main campus** of the University of Warwick (CV4 7AL). On Google Maps, it appears as “**Scarman — Warwick Conferences**”. Buses **11, 12X, 14/14A** from Coventry train station (5 min walk from the station to the Rail Station Bridge bus stop) or **U1, U1A, 11** from Leamington Spa (5 min walk from the train station to the Parish Church bus stop). Before heading from Warwick University Interchange to the venue, admire the “**Days of Judgement, Cat 1**” sculpture at the lawn next to the bus stop (marked by a star on the map).

Instructions for oral presenters

Please bring your presentation on your laptop – we will have a HDMI connection and a USB-C adapter available. If you prefer not to bring a laptop, then please send your presentation to Tom Sykes (thomas.sykes@warwick.ac.uk) in advance. You are very welcome to test your laptop connection in a break prior to your talk.

Please prepare your presentation to be **10 minutes at most**. Timings will be strictly policed, since we have hardly any slack in the schedule so as to accommodate as many talks as possible. Session chairs will give you a warning when there is 2 minutes remaining in your talk.

Following your presentation, there will be 2 minutes for questions and discussion. There is no changeover time; **during the question period, we ask the next presenter to come to the front and plug in their laptop** so they are ready to start as soon as the questions are over.

Instructions for poster flash talk presenters

This type of talk is a new innovation for MFMM, given the large number of abstracts we received and a wish to give researchers (especially ECRs) as much visibility as possible. Feedback, both from presenters and the audience, will be welcome!

Flash talks will be **3 minutes each**, with no time for questions. Any questions that the audience have can be used to kick start the discussion at the corresponding poster immediately after the flash talks. All flash talk presenters will have a poster (perhaps there is a market for a “short talk” with no poster at future MFMM meetings?).

Flash talks will be limited to **2 slides** (including everything) and **must be sent to Tom Sykes** (thomas.sykes@warwick.ac.uk) on or before **Friday 12th June**, so that they can be compiled into a single slide deck. The final running order will be shared by email with all flash talk presenters as soon as all slides are received. Please be ready to start talking immediately after the previous presenter has finished.

Remember, the purpose of your flash talk is to whet people’s appetite for your research, rather than give all details, which can be shared at your poster. Example flash talk slides will be shared with presenters via email. There is more general flash talk advice available [online](#).

Instructions for poster presenters

Please bring your poster along with you on the day – there is no need to submit it in advance. A1 or A0 size is recommended. There is a dedicated poster session in the afternoon (after the flash talks), but feel free to mount your poster in the morning on the boards, which will be in the same room as the talks. We will have mounting hardware, should you need.

Schedule

Start time	Activity
09:15	Welcome remarks
09:20	Session 1: Five 10+2 minute talks (Chair: Yongmann Chung)
09:20	Warwick Spotlight Fluids Proposal, led by Michael Auinger
09:32	Adam Preece (ARA)
09:44	Rebecca Lewis (Aston University)
09:56	Arjun Shergill (University of Birmingham)
10:08	Jialin Su (Loughborough University)
10:25	Break
10:55	Session 2: Six 10+2 minute talks (Chair: Svetlana Aleksandrova)
10:55	Mohammad Tavakol Sadrabadi (WMG, University of Warwick)
11:07	Humberto Medina (University of Nottingham)
11:19	Zhong-Nan Wang (University of Birmingham)
11:31	Bonanza Sebastian (Coventry University)
11:43	Andrew McMullan (Aston University)
11:55	Dave Smith (University of Birmingham)
12:10	Group photo
12:15	Lunch
13:30	Session 3: Four 10+2 minute talks (Chair: Christopher Davies)
13:30	Vasit Sirilapanan (Kasetsart University/ University of Warwick)
13:42	Richard McNair (University of Nottingham)
13:54	Eleanor Barton (Aston University)
14:06	Mitya Pushkin (York University)
14:20	Poster flash talks (Chair: Tom Sykes)
	Risa Romy, Jai Gill, Yanji Zhu, Hibah Saddal, Zekun Ma, Ethan Warman/Lois Martin, Sarah Bhutta, Will Biddles
14:45	Poster session
15:25	Session 4: Five 10+2 minute talks (Chair: Chandan Bose)
15:25	David Sibley (Loughborough University)
15:37	Svetlana Aleksandrova (University of Leicester)
15:49	Yukesh Karki (University of Birmingham)
16:01	Zhaoxin Ren (Swansea University)
16:13	Robert Kerr (Maths, University of Warwick)
16:25	Break
16:40	Session 5: Four 10+2 minute talks (Chair: Petr Denissenko)
16:40	Efe Guner (Engineering, University of Warwick)
16:52	Tanmay Agrawal (University of Oxford)
17:04	Dongdong Xu (University of Leicester)
17:16	Ugyen Wangchuk (Loughborough University)
17:30	Close

Session 1

Talk 1: Michael Auinger (*University of Warwick*)

Spotlight Fluids Proposal – Framework and call for involvement

Fluid flow is a very interdisciplinary field of research. As part of this broad community, our core team has come together and prepare a proposal for an interdisciplinary research proposal, spanning activities in the area of engineering, business development and interdisciplinary sciences. We would therefore use this opportunity to present the initial framework of the proposal and encourage academics to contribute to this exciting activity.

Talk 2: Adam Preece (*ARA – Aircraft Research Association*)

What, why & how of the ARA high-speed reference model

ARA has designed, manufactured and tested a generic wind tunnel model representative of a 5th generation fighter aircraft suitable for the ARA transonic wind tunnel. This presentation will cover why ARA need this model, what the model is, including its design philosophy, and how the model has and will be used to help validate and improve processes. We will also dive into the CFD and wind tunnel results, including comparisons between pressure taps and overall forces and moments. Lastly, we will provide an overview of future developments.

Adam is presenting the work of Dominic Costagliola-Ray and Miguel Aresta.

Talk 3: Rebecca Lewis (*Aston University*)

Experimental flow visualisation of impeller paddles in a bioreactor using PIV

Large quantities of stem cells are required in applications such as cultivated meat and regenerative medicine. Stirred-tank bioreactors are being relied on to improve the efficiency of cell culture to reach demand. Due to the sensitive nature of stem cells, the stirring speed used should minimise shear stresses, which damage the cells, while homogenising the environment to produce a uniform cell product. This research visualises the influence of impeller paddle geometries on velocity fields and shear stresses to identify the current challenges in bioreactor design for optimal cell growth.

Planar particle image velocimetry experiments were performed on a 1:1 scale model of a commercial 100 mL ‘Bellco spinner flask’ cast from Sylgard 184, with a refractive index matched Newtonian working fluid. A double-pulsed laser was used to illuminate a vertical plane through the centre of the model and the flow behaviour with a commercially available impeller was compared with an alternative holed paddle design at three stirring speeds relevant to laboratory operation of the spinner flask (20, 30 and 40 rpm).

The results demonstrate how impeller paddle geometries affect cell suspension and culture homogeneity at different stirring speeds. They also show that using the novel paddle design creates regions of low shear stress that are less damaging for cells. The knowledge gained will inform bioreactor design for specific cell types during the scale-up of cell culture,

aiding production of greater numbers of cells with the desired qualities for their intended application.

Talk 4: Arjun Shergill (*University of Birmingham*)

GPU-accelerated simulations of butterfly wing-wake interactions

The flight of insects has attracted significant research attention due to their remarkable manoeuvrability at low Reynolds numbers, enabled by employing unsteady vortex dynamics and specialised wing morphology. Of these, butterflies are particularly interesting: They are able to change the aspect ratio and wing sweep angle of their flexible, partially overlapped wings during flight, and different species use distinct wing shapes that produce unique aerodynamic behaviours. This work models three butterfly species with rigid wings – the Monarch, Swallowtail and Indian Leaf – using a GPU-accelerated Immersed-Boundary Method combined with an Overset Mesh. This approach allows one to capture high-fidelity vortex structures efficiently, avoiding the high cost of traditional mesh methods such as Adaptive Mesh Refinement (AMR) and overcoming the limitations of CPU-based solvers. Dual parallelism with OpenMP and OpenACC further improves the performance. Simulations reveal a consistent formation of Leading-Edge, Wing-Tip, and Trailing-Edge vortices. The swallowtail maintains an attached leading-edge vortex (LEV), while the Monarch and Indian Leaf show inboard LEV detachment due to their forward-swept forewings. This aligns with previous research. Overall, the present study demonstrates that GPU-accelerated IBM-Overset methods can accurately resolve complex flapping-wing problems. Future work will incorporate wing flexibility, body motion, and large-scale multi-body simulations.

Talk 5: Jialin Su (*Loughborough University*)

Development of multi-physics capabilities in spectral/hp element CFD

Given the general trend towards scale-resolving simulations to achieve new understanding in challenging fluid mechanics problems, high-order methods continue to gain further traction owing to their favourable numerical properties and alignment to modern computing architectures. However, in contrast to its increasing popularity in single phase flows, the use of high-order CFD in multi-physics applications is relatively limited. The spectral/hp element platform Nektar++ is a general purpose CFD package for unstructured grids, which makes it a suitable base solver for complex engineering configurations. To leverage its numerical efficiency, a multi-phase flow solver utilising a conservative level set formulation and a prototype diffusion flame combustion solver based on tabulated chemistry are developed within the Nektar++ framework. These are validated with various canonical test cases. Further work is ongoing to improve their numerics and modelling capabilities.

Session 2

Talk 1: Mohammad Tavakol Sadrabadi (WMG, *University of Warwick*)

AI augmented physics-based model for prediction of wildfire propagation

This talk presents an AI-augmented physics-based framework for improving wildfire propagation prediction to support emergency response and decision-making. The approach enhances existing decoupled wildfire spread models by integrating artificial intelligence, UAV swarm sensing, computational fluid dynamics, and deep learning. In particular, the framework uses sparse wind measurements collected by UAV swarms at flight altitude to estimate high-resolution near-surface fire-driven wind fields, which are then supplied to physics-based wildfire models to better capture fire–wind interactions. By continuously updating the model with reconstructed wind fields, the proposed system aims to improve prediction accuracy and reduce uncertainty in operational wildfire forecasting.

Talk 2: Humberto Medina (*University of Nottingham*)

Introducing XCALibre.jl: A new XPU accelerated CFD solver with native AI integration

Developed at the University of Nottingham, XCALibre.jl is a new, open-source finite-volume computational fluid dynamics (CFD) framework written in Julia. It addresses the dual challenges of heterogeneous hardware and data-driven modeling by providing seamless performance portability across diverse processing units (XPUs). The solver natively supports NVIDIA, AMD, and Intel GPUs and multi-core CPUs.

Crucially, XCALibre.jl bridges the gap between traditional numerical methods and artificial intelligence. Operating natively within the Julia ecosystem, it enables researchers to effortlessly integrate custom machine learning models (e.g., via Flux.jl) directly into the CFD numerics. This eliminates traditional language-bridging overhead, allowing for rapid prototyping of data-driven boundary conditions, neural-network-augmented turbulence closures, automated optimisation workflows, etc.

XCALibre handles 2D and 3D unstructured meshes, turbulence modeling (RANS and LES) for incompressible and compressible flows. This presentation will introduce the solver’s highly extensible architecture, showcase its capabilities, and highlight a roadmap of upcoming features to accelerate next-generation fluid mechanics research.

Talk 3: Zhong-Nan Wang (*University of Birmingham*)

Learning nonlinear dynamics of turbulent flows for noise generation

The noise generation by turbulence is highly related to coherent structure, but linear model can produce the on-average dynamical behaviour of the coherent structures but significantly underpredict the noise radiation. This indicates that nonlinearity plays a key role in generating noise. In this research, we identified the nonlinear interactions of coherent structures that underpins the noise generation by inferencing a reduced-order model (ROM) from high-fidelity simulation data. To enhance the accuracy and interpretability of ROM, the physical constraint has been applied to preserve the energy transfer in turbulence energy cascade,

and a closure model was developed to account for the interactions with untruncated parts. In addition to the deterministic interactions between coherent structures, a stochastic forcing is formulated to account for the effects of fine-scale turbulence. The derived ROM can well represent the nonlinear dynamics for a mixing layer, compared to high-fidelity simulation data, and are also able to separate the contribution from linear and nonlinear parts. The acoustic field was obtained by solving the Acoustics Perturbation Equation with reduced-order modelled sources, offering insights into the relationship between the nonlinear dynamics of coherent structures and noise generation.

Talk 4: Bonanza Sebastian (*Coventry University*)

The effect of viscosity on the streaming instability in protoplanetary disks

In disks where planets form, dust and gas orbit the central star at slightly different speeds. This speed difference can trigger an instability, known as the streaming instability (Youdin & Goodman 2005), that causes dust to clump together, representing a key step toward building planets. Previous studies of this instability have often focused on inviscid models. In our work, we include viscosity to investigate whether it acts as a meaningful stabilising mechanism and to better understand its role in the process.

Using linear stability analysis, we study how viscosity affects the growth of this instability across a range of particle sizes and dust-to-gas ratios. We find that viscosity shrinks the region of instability in parameter space and reduces peak growth rates, and that beyond a critical viscosity level, the instability is suppressed. We then apply this analysis to twelve real observed disks at orbital distances between 10 and 100 AU, using local disk conditions derived from observations, to assess where and under what conditions the instability can realistically operate. By converting our results into physical grain sizes, we identify the particle sizes at which the instability is most vigorous in each disk, finding values consistent with millimetre-sized particles, a range directly accessible to current telescope observations.

In this meeting, I will present these results and discuss their implications, particularly how viscosity may play a more important role in planet formation.

Talk 5: Andrew McMullan (*Aston University*)

LES of the variable density mixing layer

A widely-accepted growth and entrainment model of the turbulent mixing layer is based on the assumption that the normalised spacing of the large-scale turbulent vortex structures can be represented by a geometric progression, which is a function of the velocity ratio parameter, and invariant of the density ratio, between the two streams. We perform a series of Large Eddy Simulations of the plane turbulent mixing layer for a range of values of the velocity ratio parameter, and the density ratio. The large-scale Brown-Roshko structures present in these simulations are analysed, and the dependency of the normalised structure spacing on the velocity ratio parameter and density ratio is assessed. The results are discussed in the context of the aforementioned entrainment model.

Talk 6: Dave Smith (*University of Birmingham*)

CNearest: a toolkit for Stokes flow

I will describe the package CNearest, which implements a method for solving problems in very viscous fluid dynamics, in the limit where inertia is negligible relative to viscosity. This system is particularly relevant in microscale biological systems and colloids; one specific application is the calculation of the diffusion tensor of a specified structure. The underlying algorithm is the nearest-neighbour implementation (Smith, J. Comput Phys 2018) of the method of regularised stokeslets (Cortez, SIAM J Comput Sci 2001; Cortez et al. Phys Fluids, 2005); this approach enables efficient and accurate calculations requiring only surface point discretisations rather than surface or volumetric meshes and avoids singular integrals. It is a feature-complete reimplement and further development of the key test cases in Smith (2018). The package also serves as an extensible library that can be utilised with the user's own surface discretisation to calculate the force and moment on any given three dimensional body undergoing rigid body motion, and provides a building block to model flexible swimming bodies.

Session 3

Talk 1: Vasit Sirilapanan (*University of Warwick, Kasetsart University*)

Hemodynamic effects of local coil packing density on aneurysm recurrence: A patient-specific CFD study

Endovascular coiling reduces aneurysm rupture risk, yet flow recanalisation can persist even at high global packing densities, and the governing haemodynamic mechanisms remain unclear. This study investigates the effect of local coil packing density (PD) on intra-aneurysmal flow using patient-specific non-Newtonian CFD. An aneurysm model reconstructed from computed tomography images was virtually coiled across seven PD levels, and the Navier-Stokes equations were solved using Star-CCM+ and high-performance computing clusters. Increasing PD produced progressively stronger flow suppression, with a pronounced first-coil effect and up to 75% velocity reduction at a PD of approximately 30%. Haemodynamic indicators of thrombus-prone regions were quantified using a low and oscillatory wall shear stress index. Passive scalar transport, comparable to clinical dye visualisation, was modelled to quantify blood residence time, relevant to thrombus-induced occlusion and treatment success. These results demonstrate that aneurysm occlusion and recurrence cannot be reliably predicted from global packing density alone, highlighting the role of local haemodynamic heterogeneity and CFD simulations in clinical applications.

Talk 2: Richard McNair (*University of Nottingham*)

Compaction in poroelastic media with deformable, impermeable boundaries

It has been shown that water driven vertically through a packed bed of hydrogel beads can cause the medium to compact, reducing porosity such that the flow rate through the medium plateaus to a constant as the pressure head diverges to infinity. This is caused by a balance between the decreasing permeability and the increasing pressure gradient across the cylin-

der. We investigate a two-dimensional extension of this problem in a tall, thin, axisymmetric porous cylinder where the radial boundaries are enclosed by a deformable, impermeable membrane with different material properties to the porous medium itself.

We exploit the small aspect ratio limit to find the leading-order flow rate and deformation as functions of the driving fluid pressure, and the material properties of both the porous medium and the bounding membrane. Our model predicts how the deformable membrane augments compaction, and allows the flow rate to diverge with applied pressure for a relatively wide range of porous media. The flow-rate plateau still exists in a region of the parameter space where compaction of the porous medium occurs without significant expansion of the bounding membrane. This study contributes to a better understanding of flow and deformation in porous media including soft biological tissues, such as the human placenta, which can be aptly modelled as poroelastic media.

Talk 3: Eleanor Barton (*Aston University*)

Particle Tracking Velocimetry of in vitro graft-treated aortic models

Cardiovascular disease presents in the aorta, as aneurysms or dissections and can be surgically repaired by replacing diseased sections of the vessel with vascular grafts, which results in altered haemodynamics. This study explores the impact of rigid branched vascular grafts on pulsatile flow dynamics, to recognise flow behaviour relevant to long-term performance.

Phantom geometries were derived from product dimensions representative of graft-treated aortas. The phantoms were cast from Sylgard 184, and experiments used a Newtonian blood analogue with a matched refractive index ($n = 1.414$). A pulsatile pump provided the aortic waveform, resistance was applied to each outlet to replicate physiological flow distribution, and flow behaviour was captured using particle tracking velocimetry. Results demonstrate that at Reynolds numbers of 3400 (peak systole), 140 (leakage) and 225 (reverse flow), geometric features such as differences in branch angles and diameters affect flow division, recirculation zones, and local shear regions. This in turn affects the secondary flow structures in the aortic arch in each graft. These findings provide novel insight into blood flow patterns in graft-treated aortas identifying regions of low velocity and low shear rate that may negatively impact re-endothelialisation, a key factor in long-term graft performance.

This work enhances understanding the effects of pulsatile flow in graft-treated aortas, informs branched vascular graft design and supports computational fluid dynamics validation for regulatory approval of vascular grafts.

Talk 4: Mitya Pushkin (*University of York*)

Topological transitions in microswimmer flows in confinement controlled by elongation and friction

Interactions of microswimmers in confinement are not well understood, despite being important for biological and technological applications. It has been shown that, far from mere dampening of the flows, confinement leads to diverse flow structures around microswimmers [1-4].

In this work, we present analytical solutions for the flows around elongated microswim-

mers in confinement. We show that the interplay between confinement, swimmer elongation, and boundary friction gives rise to cascades of topological transitions in the flows.

[1] Pushkin and Bees., 2016. Bugs on a Slippery Plane: Understanding the Motility of Microbial Pathogens with Mathematical Modelling. In *Biophysics of Infection* (pp. 193-205).

[2] Jeanneret, Pushkin, and Polin, 2019. Confinement enhances the diversity of microbial flow fields. *Physical Review Letters*, 123(24), p.248102.

[3] Gilpin, Prakash, and Prakash, 2017. Vortex arrays and ciliary tangles underlie the feeding–swimming trade-off in starfish larvae. *Nature Physics*, 13(4), pp.380-386.

[4] Mondal, Prabhune, Ramaswamy, and Sharma, 2021. Strong confinement of active microalgae leads to inversion of vortex flow and enhanced mixing. *Elife*, 10, p.e67663.

Poster presentations

Many of the following posters will be introduced by the presenter in flash talks that immediately precede the poster session.

Poster: James Bell (*University of Nottingham*)

The structure of a drain vortex in a rotating system

Vortices in rotating systems are widely observed in both natural and industrial settings and are of particular relevance to geophysical flows such as mesoscale ocean eddies and atmospheric vortices. Such systems can often be described as having low Ekman and Rossby numbers, meaning the Coriolis term dominates over the viscous and nonlinear effects respectively.

An asymptotic solution for an axisymmetric drain vortex in a rotating system has been developed for small Rossby and Ekman numbers. The overall flow structure predicted matches observed found in experiments and it agrees quite well quantitatively with limited data from previous studies, which have few data points and do not completely match the conditions of the theory.

The work presented here reports laboratory experiments run specifically to provide detailed comparison with these asymptotic solutions. Velocity fields are measured using particle image velocimetry in experiments that more closely match the theoretical conditions. Initial results show qualitative agreement, while measurable quantitative discrepancies in velocity measurements have been observed and are the subject of ongoing investigation.

Poster: Risa Romy (*University of Warwick*)

Arteriovenous fistula health monitoring using ultrasound imaging and Computational Fluid Dynamics

One of the most common ways of managing kidney failure is haemodialysis which involves connecting patients to a machine that performs the function of the kidneys. The arteriovenous fistula (AVF) is the most popular form of vascular access, used to facilitate this treatment which requires blood flow rates of around 300 to 500 mls/min [1]. Patients undergo surgery to connect the artery to the vein resulting in increased venous blood flow. The new

AVF matures over a period of 4-6 weeks to allow venous remodelling and thickening in response to the increased flow rates to support regular cannulation for dialysis [2]. Unfortunately, there is a high failure rate with around 40% of failures within the first year [3], most commonly caused by stenosis, a narrowing often resulting from intimal hyperplasia (IH). Currently, the complex AVF haemodynamics and pathology-causing mechanisms are poorly understood, in particular, how the haemodynamics contribute to stenosis development. Ultrasound (US) imaging is a common tool used to aid diagnosis and management, however, it only offers limited information. This is where computational fluid dynamics (CFD) is useful as it enables the detailed study of complex AVF haemodynamic characteristics specific to patients [4]. The aim of this research is to use ultrasound imaging for patient-specific AVF segmentation and computational fluid dynamics study to observe the complex haemodynamic patterns for AVF health monitoring.

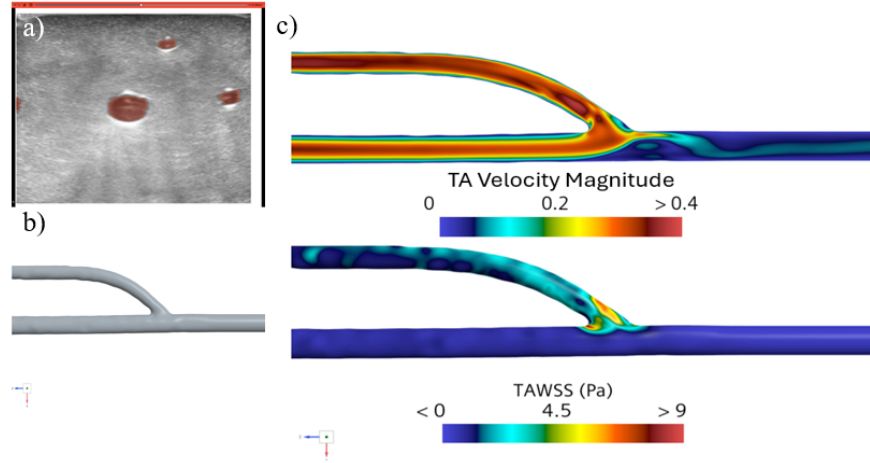


Figure 2: (a) Segmented US image. (b) 3D reconstructed geometry. (c) Time-averaged velocity (top) and WSS magnitude (bottom).

While US imaging will be used in future work for blood vessel segmentation to capture accurate patient-specific AVF geometries, three-dimensional (3D) reconstruction from US images acquired using a two-dimensional (2D) hand-held probe remains highly challenging (Figure 2 (a)). In this study, computed tomography (CT) images were used as a benchmark for subsequent comparison with US-based reconstructions. Figure 2 (b) shows the 3D geometries reconstructed from CT scans. Figure 2 (c) presents the time-averaged velocity magnitude and wall shear stress magnitude for a CT-segmented geometry. The connection between a high-velocity, high-pressure arterial vessel and a low-velocity, low-pressure venous vessel often induces severe flow disturbances. Recirculation zones are frequently observed along the inner curvature of the arch. Previous studies suggest that non-physiological flow patterns play a pivotal role in the long-term function and health of AVFs, with evidence indicating that they are key contributing factors to IH and stenosis [4,5]. The next stage of this research project will involve using 3D-reconstructed US images to create patient-specific 3D AVF models. This work is being undertaken in collaboration with Siemens Healthineers, USA. The entire process can be automated to enable rapid segmentation and CFD analysis,

facilitating the efficient processing of a larger number of cases. US image-based, patient-specific CFD has the potential to provide a non-invasive approach to monitoring complex AVF haemodynamics.

- [1] Lok C.E. and Foley R. Vascular Access Morbidity and Mortality: Trends of the Last Decade. *American Society of Nephrology*. 2013; 8: 1213-1219.
- [2] Hindle P., Ritchie J., Griffiths G., and Howd A. *The Journal of Vascular Access*. 2011;12(20), 127-129.
- [3] Salih S.S.M., Mohamed K.O., Mohamedali A.O.O., et al. *Patient Saf Surg*. 2025;19(1):24. Published 2025 Sep 2. doi:10.1186/s13037-025-00449-9.
- [4] Bartlett M., Bonfanti M., Diaz-Zuccarini V., and Tsui J. *Reviews in Cardiovascular Medicine*. 2024, 25(1), 35.
- [5] Fitts M.K., Pike D.B., Anderson K., and Shiu Y.T. *Open Urology and Nephrology Journal*. 2014;7(Suppl 1 M5):33-44.

Poster: Zekun Ma (*University of Warwick*)

The effect of temporal acceleration on uniform momentum zones of turbulent channel flow

Driven by the demand for low noise, safe, and low carbon aviation, tiltrotor aircraft offer a promising solution for future urban air mobility. A tiltrotor combines the capability of vertical take-off and landing (VTOL) with the advantage of high-speed cruise flight [1], with earlier practical applications represented by the Bell-Boeing V-22. The aerodynamic design of tiltrotor aircraft remains challenging. One key issue is the accurate prediction of downforce during hover. Owing to the highly twisted, low-aspect-ratio blades and high disk-loading design, the relatively thick inboard blade sections may exhibit delayed stall as a result of three-dimensional and centrifugal effects [2]. The prediction of the flow field and aerodynamic characteristics in conversion mode between hover and cruise remains challenging. During the conversion, the rotor wake becomes highly distorted, and the blade loading exhibits strong unsteadiness, which makes blade control and aerodynamic performance prediction more difficult [3]. In this study, computational fluid dynamics (CFD) simulations are performed for the Vertical Aerospace VX4 model. The objective is to investigate the physics and explore novel design concepts for the Vertical Aerospace VX4 model.

Figure 3(a) shows a schematic of the tilting motion. The tilting motion is centred at O' , while the propeller rotation is centred at O'' . As shown in Figure 3(b), the aircraft has eight propellers in total. The front four propellers can tilt to enable conversion between hover and cruise. Figure 3(c) shows the wake structures visualised using the Q-criterion, with the colour indicating the velocity magnitude.

1. Potsdam, M. A. & R. C. Strawn 2005 CFD simulations of tiltrotor configurations in hover. *Journal of the American Helicopter Society* 50, 182–94.
2. Narducci P, Ju Liu, A J Wells, F J Mobley & R J Mayer 2024 CFD simulations of a hovering tiltrotor in ground effect. *AIAA SCITECH 2024 Forum*, AIAA 2024–1115.
3. Zhang, Y. L. Ye & S. Yang 2015 Numerical study on flow fields and aerodynamics of tiltrotor aircraft in conversion mode based on embedded grid and actuator model. *Chinese Journal of Aeronautics* 28, 93–102.
4. Archer Aviation Inc. 2026 *Midnight aircraft*. <https://archer.com/aircraft>. Accessed 26 May 2026.

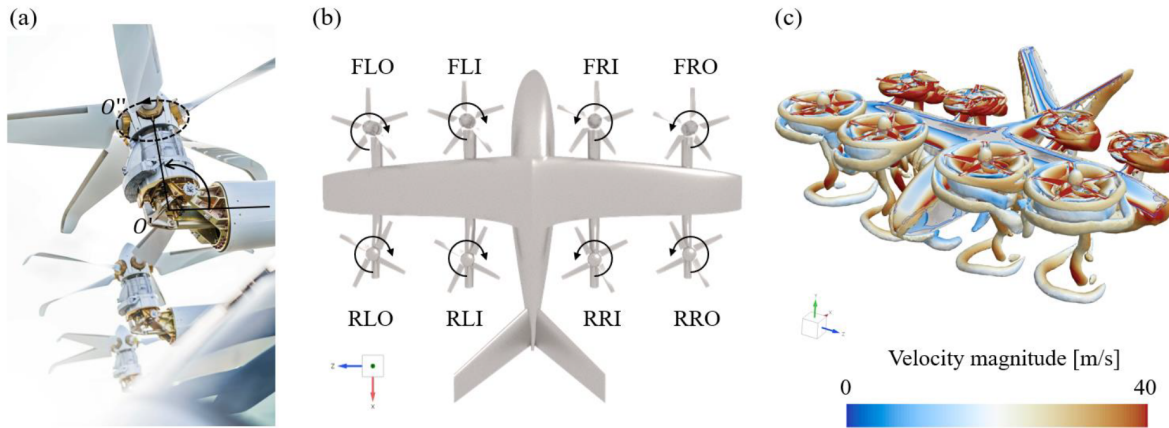


Figure 3: (a) Schematic of the tilting motion, reproduced from [4]. (b) CAD model used in the present simulation. (c) Wake structures, visualised using the Q-criterion and coloured by velocity magnitude.

Poster: Sarah Bhutta (*University of Warwick*)

Modelling and optimisation of battery electrode slurry preparation process by extrusion

Batteries are used in applications ranging from electric vehicles to portable electronics. A critical step in their manufacturing is the preparation of electrode slurry, where active materials, conductive additives, and binders are dispersed in a solvent to form a homogeneous mixture. Conventional batch mixing processes often struggle to maintain consistent slurry quality due to limited control over mixing dynamics, resulting in uneven particle dispersion, high energy consumption, and challenges in scaling production. These limitations can adversely affect electrode microstructure, electrochemical performance, and overall manufacturing efficiency.

This work presents a predictive modelling framework for lithium-ion battery slurry preparation using twin-screw extrusion (TSE) as a continuous mixing approach. The proposed framework aims to improve understanding and control of slurry mixing behaviour while enabling scalable and more energy-efficient manufacturing processes.

A multi-scale methodology is employed, integrating Computational Fluid Dynamics (CFD) simulations with experimental validation using lab-scale extrusion trials. The modelling framework captures slurry flow behaviour and mixing characteristics within the extruder while systematically investigating key parameters such as screw speed, solid content, residence time distribution (RTD), and particle size distribution.

The study provides insights into the relationship between extrusion processing conditions and slurry homogeneity, highlighting how process parameters influence particle dispersion and mixing efficiency. The developed modelling framework offers a generalisable tool to support process optimisation, improve manufacturing consistency, and accelerate the adoption of continuous processing technologies for lithium-ion battery electrode production.

Poster: Jai Gill (*Aston University*)

Stability analysis and saturated subcritical states in the Taylor-Couette-Poiseuille system

By applying stability analysis to the Taylor-Couette-Poiseuille system, specifically in the narrow-gap limit, we examine the existence of saturated, non-linear, subcritical states. Utilising an RK4-based solver, linear analysis allows us to identify a threshold for the stability of the system to infinitesimal perturbations for the control parameters: Omega for rotation, the pressure Reynolds number and Couette Reynolds numbers. In order to capture the finite-amplitude and structure of the higher dimensional vortices, we employ a non-linear spectral method below the linear stability curve to determine whether a subcritical bifurcation takes place due to finite-amplitude perturbations. Our results confirm the existence of finite-amplitude, saturated, non-linear states below the linear stability curve, indicating the presence of subcritical bifurcations.

Poster: Will Biddles (*University of Warwick*)

Investigation of ground effect on a bluff body using high fidelity LES

The Ahmed body is a bluff body used to study ground vehicle wake dynamics. It is used instead of complex real geometries as it is easier to model numerically, and the model's geometries is designed so that it creates the same wake features found in real vehicle wakes.

The majority of research conducted using the Ahmed body has been conducted using a stationary ground plane. In order to better capture the true conditions experienced by ground vehicles, the present study implements a sliding ground condition to mimic real road conditions. It is important to understand the impact that the ground condition has on the flow around the body. This study will present a comparison between the two different conditions, the impact the ground effect has on the coherent structures, the forces the body experiences, and how this compares to previous studies [1].

Large eddy simulations (LES) are conducted on Ahmed body with a slant angle of 25° . The simulations are performed using the spectral element method (SEM) solver *Nek5000* where a hexahedral mesh is used. An LES model is employed using a high pass filter [2]. $\mathbb{P}_n - \mathbb{P}_{n-2}$ discretisation is used, with a polynomial order of $n = 7$. A third order backwards difference time scheme is utilised in this study. The Reynolds number, based on the height of the Ahmed body is, $Re_h = 14,800$.

A uniform inlet and turbulent outflow conditions are implemented [3]. Periodic boundary conditions are applied to the sides, and a symmetry (slip) condition is applied to the roof of the domain. No-slip conditions are applied to all surfaces of the body, and the ground for the stationary ground case. A tangential velocity is applied to the ground for the sliding ground condition. Previous studies have observed up to a 10% decrease in drag coefficient from the no-slip to sliding case [1]. Results in this study indicate almost a 4% reduction in drag and approximately a 19% reduction in lift.

Results from time averaged streamwise velocity profiles from both ground condition simulations, and results from a water tank test [4], are compared. No slip LES results show reasonable agreement with the water tank experiment. Both conditions have a disturbance near the trailing edge of the underbody, with the sliding condition having a significantly larger disturbance. Above the body, at the top of the slanted face, near the surface there is

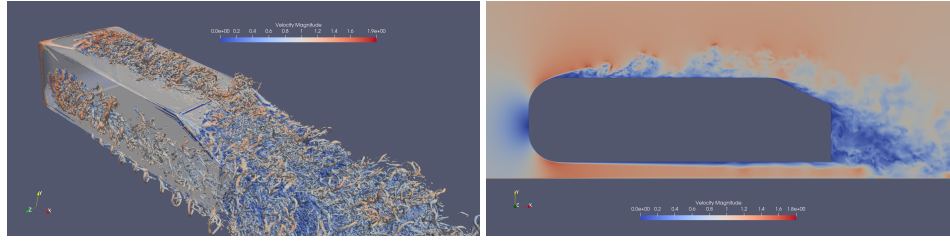


Figure 4: Left: $\lambda_2 = -25$. Right: Velocity profile along symmetry plane.

negative or zero velocity, due to the presence of a recirculation bubble as a result of flow separation and reattachment.

1. S. Krajnović and L. Davidson. Influence of floor motions in wind tunnels on the aerodynamics of road vehicles. *Journal of Wind Engineering and Industrial Aerodynamics*, 93:677–696, 2005.
2. S. Stolz, P. Schlatter, and L. Kleiser. High-pass filtered eddy-viscosity models for large eddy simulations of transitional and turbulent flow. *Physics of Fluids*, 17:065103, 2005.
3. S. Dong, G.E. Karniadakis, and C. Chrysosostomidis. A robust and accurate outflow boundary condition for incompressible flow simulations on severely-truncated unbounded domains. *Journal of Computational Physics*, 261:83–105, 2014.
4. T. Tunay, B. Sahin, and H. Akilli. Experimental and numerical studies of the flow around the Ahmed body. *Wind and Structures*, 17:515–535, 2013.

Poster: Yanji Zhu (*University of Warwick*)

A coupled Level-Set and VOF method for modelling insoluble surfactant transport with high Peclet numbers

The transport behaviour of insoluble surfactants occurs exclusively at the phase interface, posing challenges for simulation within the framework of computational fluid dynamics (CFD), particularly regarding monotonicity, numerical diffusion and dissipation. Current Eulerian models primarily focus on diffusion-dominant problems with low Peclet numbers, whereas the diffusion of common surfactants is very limited. Based on an existing open-sourced Basilisk code, an Eulerian CFD model is developed to simulate insoluble surfactant transport with realistic Peclet numbers (small diffusion coefficient). The code is validated through a range of physical scenarios, including advection, diffusion, bubble expansion, and linear shear flow.

Poster: Hibah Saddal (*University of Birmingham*)

Spatio-temporal gust response of multi-segment passive morphing foils with variable flexibility

This study computationally investigates the coupled fluid-structure interaction response and wake dynamics of multi-segment passive morphing foils with variable flexibility. Particular focus is on understanding the effects of transient acceleration and spatially varying verti-

cal gusts on the aerodynamic performance in the low Reynolds number regime. The fluid flow is governed by the incompressible Navier-Stokes equations and solved using the finite volume method in OpenFOAM, while the structural response is simulated using the finite element based solver CalculiX. Two-way fluid-structure interaction coupling achieved using the Interface Quasi-Newton Inverse Least Squares (IQN-ILS) acceleration scheme. Radial basis function interpolation is employed for data mapping across the fluid-solid interface. The study examines the influence of key governing parameters, including the dimensionless bending rigidity and density ratio, on wake behaviour and aerodynamic performance. The results indicate that passive morphing mechanisms can mitigate the spatio-temporal effects of gust disturbances, leading to improved aerodynamic efficiency. These findings provide insight into vortex-dominated flow physics, directing benefitting the design of futuristic micro air vehicles and unmanned aerial vehicles operating in gust-laden environments.

Poster: Ethan Warman/Lois Martin (*University of Birmingham*)

Wake dynamics of pitching swept wings at low Reynolds numbers

The unsteady wake dynamics of sinusoidally pitching finite-aspect-ratio wings are investigated numerically at a chord-based Reynolds number of $Re = 1000$. High-fidelity three-dimensional simulations are conducted using the incompressible pimpleFoam solver within OpenFOAM. The study examines sweep angles of $\pm 30^\circ$, $\pm 45^\circ$, and $\pm 60^\circ$, semi-aspect ratios of 2 and 4, and reduced frequencies of 0.5 and 1.0. The pitching amplitude is maintained at $\pm 25^\circ$. The evolution of leading-edge vortex (LEV) topology, tip vortex (TV) interactions, and wake organisation is characterised using Q-criterion isosurfaces, spanwise vorticity, surface pressure distributions, and vortex circulation tracking. The results demonstrate a fundamental asymmetry between sweep directions. Backward sweep promotes spanwise vorticity transport toward the wing root, stabilising the LEV and facilitating its coalescence with the tip vortex. This process results in a characteristic arch-vortex system. In contrast, forward sweep drives circulation inboard and induces early LEV fragmentation and loss of spanwise coherence, particularly at higher sweep angles and lower frequencies. Stall initiation progresses from the tip to the root in BSWs and from the root to the tip in FSWs. Increasing the aspect ratio amplifies midspan effects. A higher reduced frequency intensifies LEV formation, advances vortex breakdown, and increases wake disorder. These findings indicate that sweep direction fundamentally governs vortex stability, force production, and three-dimensional wake organisation in oscillating low-Reynolds-number wings. The results have direct implications for the aerodynamic design of micro-aerial vehicles and bio-inspired propulsors.

Session 4

Talk 1: David Sibley (*Loughborough University*)

Coexisting multiphase and interfacial behaviour of ouzo

Multi-component liquid mixtures can be both complex and fascinating, with some systems being amenable to simple experimentation at home, giving valuable insight into fundamental aspects of bulk and interfacial phase behaviour. One particularly interesting mixture is the popular drink ouzo, which has charmed both the general public and scientists by virtue of its ability to display spontaneous emulsification when water is added. When these two clear (& drinkable) liquids are poured into each other, a single milky-coloured liquid is formed.

In this seminar I will discuss some recent collaborative work finding the equilibrium phase-diagram for the stable liquid phases of ouzo via experiment and modelling and extensions for when the two liquid phases also coexist with a vapour phase (i.e., along a line of triple points) and within our model uncover the complex bulk phase behavior for this simple beverage. As a consequence, this leads to some interesting observations that also apply more widely, about visualising phase diagrams in ternary systems of this type. Our modelling relies on a relatively simple lattice density functional theory model but captures physical reality surprisingly well. We also examine the interfacial behaviour, connecting microscopic density functional theory results with macroscopic (Neumann) predictions for the shape of droplets at interfaces.

Talk 2: Svetlana Aleksandrova (*University of Leicester*)

Porous medium approach for modelling lung ventilation and drug delivery

Understanding airflow in the human airways is important for improving diagnosis and treatment of pulmonary disease, from assessing ventilation changes in cystic fibrosis to evaluating how effectively inhalers deliver medication in asthma. Reliable Computational Fluid Dynamics (CFD) models can support both research and clinical applications, including patient-specific analysis of ventilation and drug delivery.

A major challenge is that modelling the full airway tree is computationally expensive, so studies often use truncated airway geometries, focussing only on the larger conducting airways. One-dimensional approaches, such as volume-filling models, allow to model distal airways and are more computationally efficient. However, they are difficult to couple with accurate modelling of the geometrically and physically complex upper airways, which is crucial for reliable predictions. The porous medium modelling approach can help bridge this gap by enabling the larger airways to be resolved in detail while representing the influence of the smaller airways and alveolar region in a computationally practical way.

In this study, the porous medium model is used to represent the smaller airways and alveoli while coupling them to the larger conducting airways. The model is shown to represent lung ventilation well and to produce realistic flow fields and particle deposition patterns without a large increase in computational cost, while avoiding the assumptions required in truncated geometry simulations. The model also offers flexibility to include additional lower-airway effects such as hysteresis and physical lung expansion and contraction. In addition, it can reflect disease state, patient characteristics, and regional differences between

lobes and lung sections, allowing more realistic simulation of airflow and particle transport.

Talk 3: Yukesh Karki (*University of Birmingham*)

Spatio-temporal spanwise twist-induced vortex dynamics of a actively morphing wing

Natural fliers often augment their propulsive efficiency by deforming their wings during different flight conditions, with spanwise deformation playing important role in modifying aerodynamic force production through its interaction with unsteady flow structures. In the present study, the effect of actively imposed spatio-temporal twist on the aerodynamics of a finite rectangular wing is investigated using three-dimensional numerical simulations. An overset mesh framework implemented in OpenFOAM is used to simulate the prescribed deformation. The wing undergoes a ramped spanwise twist in which the root remains fixed, and the twist amplitude increases towards the tip. Different spanwise twist distributions are considered to examine how the spatial variation of deformation affects the surrounding flow field. The study investigates how imposed twist affects the formation of leading-edge vortices, tip vortices, and wake structures, and how these changes impact aerodynamic force generation and flight performance. It highlights the role of controlled spatio-temporal deformation in influencing vortex behaviour and unsteady aerodynamics.

Talk 4: Zhaoxin Ren (*Swansea University*)

Phase-transition-driven instabilities in under-expanded cryogenic hydrogen jets

Under-expanded cryogenic hydrogen jets exhibit strong shock–vortex interactions and rapid thermodynamic changes that drive complex multiphase behaviour. Using a high-fidelity CFD framework, this study investigates two coupled phase-transition mechanisms: hydrogen liquefaction and water vapour condensation. Liquefaction is shown to originate from expansion-wave-induced pressure drops and coherent vortex structures, while ambient humidity strongly influences condensation dynamics in the surrounding air. Increasing humidity enhances condensation-induced density gradients, accelerates flow destabilisation, and reduces jet penetration. The results provide new physical insight into phase-transition-driven instability mechanisms in cryogenic hydrogen jets, relevant to propulsion system design and hydrogen safety.

Talk 5: Robert Kerr (*University of Warwick*)

Spectra and structure function statistics from numerical vortex knots

Following the evolution of a patch of turbulent fluid, far from boundaries, presents challenges to both experiments and numerics. For experiments, the first problem is how to remove boundaries from the creation of a turbulent patch. And for numerics, one initial question is whether the patch is physical, or just random noise. A plausible solution for both to initialise with knotted vortices.

A new water tank experiment¹ uses the collision of several vortex rings to achieve this, with spectral results consistent with classical turbulence. This presentation will show numer-

ically how one type of vortex knot can obtain similar, but lower Reynolds number, spectra as well as other examples of classical diagnostics. The numerical innovation is to use vortex knots² that at $t = 0$ are not prone to infinitesimal instabilities, quickly generate finite-time energy dissipation, and are more compactly isolated from the domain boundaries than before³. Besides generating Kolmogorov spectra whose scaling is like that from the new experiment and are consistent with observations of isotropic, homogeneous turbulence, the diagnostics of the growth of turbulent blob can be related to the statistical properties of the classical, incompressible, Navier-Stokes equations. And in particular, should provide us with a means of comparing new mathematics for how vorticity moments and structure functions evolve together.

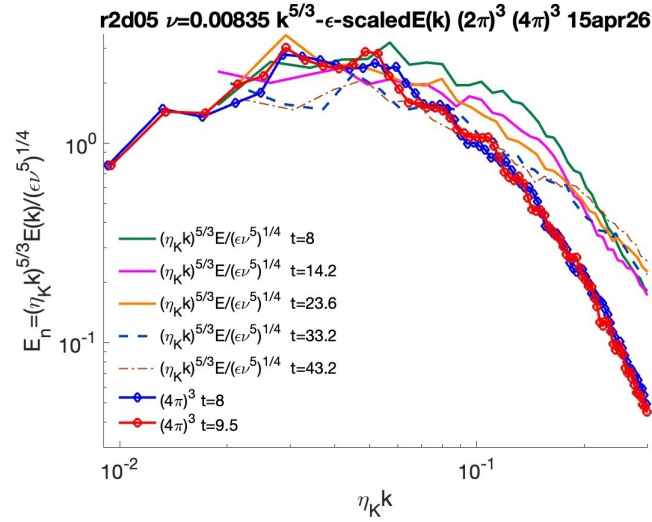


Figure 5: (a) Convergences of $k^{5/3}$ compensated, Kolmogorov-normalised energy spectra over time from two simulations with the same Rosenhead profile compact trefoil initial condition² with $p_r = 2$ and $r_o = 0.05$ with maxima of the dissipation rates $\epsilon = \nu Z$ at $t \sim 5.1$ and convergent ϵ at $t \sim 12$. The first case was done in a $(2\pi)^3$ domain with 512^3 resolution. The second used a $(4\pi)^3$ domain with 1024^3 mesh points. The $(2\pi)^3$ case maintains a $k^{-5/3}$ inertial subrange until $t = 23.6$. In the $(4\pi)^3$ domain, the inertial subrange is similar and at low- k the spectra dips, as in the experiment.

- 1) T. Matsuzawa *et al.*, Nonlinear diffusion and decay of a blob of turbulence spreading into a quiescent fluid. *Proc. Natl. Acad. Sci. U.S.A. (PNAS)* 123(7), e2526858123. 2026.
- 2) R.M. Kerr, Sensitivity of trefoil vortex knot reconnection to the initial vorticity profile. *Phys. Rev. Fluids* 8, 074701. 2023.
- 3) R.M. Kerr, Compact Navier–Stokes trefoils in large domains with finite dissipation. *J. Fluid Mech.* 1025, A3. 2025.

Session 5

Talk 1: Efe Guner (*University of Warwick*)

Drag force investigations in a rotating fluid

The drag force acting on a sphere rising under the action of buoyancy in a liquid-filled, rotating fluid is investigated experimentally. The consideration of drag forces is essential in connection with many engineering design problems, because drag affects the power required to propel objects through a fluid. Rotating liquids are encountered frequently in engineering applications or geophysics. Coriolis forces arising in a rotating fluid substantially alter the flow physics in comparison to the corresponding flow problem in a non-rotating fluid. They lead to many, often counterintuitive, effects that are entirely absent in a non rotating fluid.

This study focuses on understanding the role of inertial, Coriolis, and viscous forces and the effects of the formation of Taylor columns in a rotating fluid. The problem is governed by dimensionless parameters such as the Reynolds (Re), Taylor (Ta), and Rossby (Ro) numbers. A vertical container 72.5 cm tall and 20 cm wide is filled with water and rotated at rates of up to around 157 rpm. A sphere with a diameter of 4 cm is released at the bottom of the cylinder after an initial spin-up phase, allowing the water inside the tank to reach the state of solid-body rotation and thereby spin at the same rate as the cylinder. Results relating to the axial velocity of the sphere, and changes of the overall flow pattern for successively increasing rotation rates are presented, analysed, and discussed.

Talk 2: Tanmay Agrawal (*University of Oxford*)

Mechanism of fluid infiltration and mixing in plane jets with applications to environmental flow

Air curtains (AC) are contemporary devices that are used to minimize buoyancy-driven exchange flow in buildings. When physical barriers are impractical e.g. in cases of human and vehicular traffic, air curtains serve as an effective alternative to mitigate such exchange flows while maintaining accessibility. These devices generate a high-momentum vertical plane jet that suppresses transverse fluid exchange, thereby assisting towards preservation of indoor air conditions. Their performance is generally quantified in terms of a separation effectiveness, E , which compares the exchange flow rates with and without an air curtain. Under optimal conditions, they restrict up to 85% of outdoor fluid from infiltrating the indoor region. In the present study, we use large-eddy simulations and a Lagrangian analysis framework to investigate the spatial origin and transport mechanisms of fluid that penetrate an air curtain. We further explain the saturation of sealing effectiveness at 85% by comparing transport characteristics of an AC with those of non-buoyant jets and gravity currents. We observe that prior to the curtain establishment, most infiltration occurs via a gravity current, where particles initially near the wall are transported indoors along near-straight trajectories. Once the AC establishes, an entrainment-detrainment pathway becomes dominant. The outdoor fluid is first entrained into the turbulent core and subsequently detrained into the indoor region, either before reaching the wall or after its impingement. Present study thus highlights that it is inherent entrainment of the ambient fluid by the curtain, which unavoidably results in a sealing effectiveness of less than 100%.

Talk 3: Dongdong Xu (*University of Leicester*)

Resolvent analysis of strongly stratified flows with arbitrary equations of state

This paper investigates the optimal growth of linear disturbances in strongly stratified flows with arbitrary equations of state. A resolvent-based framework is developed that extends classical resolvent analysis to thermodynamically complex regimes, where fluid properties exhibit significant variations. An asymptotic expansion for small Mach numbers is employed to filter out acoustic waves, thereby clarifying the interpretation of the results. The optimal gain is computed over a range of wavenumbers and frequencies by performing a singular value decomposition of the resolvent operator and by defining an energy norm tailored to the problem. We discuss the different energy growth mechanisms (lift-up and Orr) and how they are affected by strong stratification. These results provide new insights into the non-modal linear dynamics of supercritical fluids – clarifying, in particular, the recent findings of Boldini et al. (PRF 2024) – while opening perspectives for their transition to turbulence and for modelling coherent structures in the turbulent regime.

Talk 4: Ugyen Wangchuk (*Loughborough University*)

Investigation of Transitional Shock Wave Boundary Layer Interaction (SWBLI) using flux reconstruction method and positivity-preserving entropy-based filtering approach

Scale Resolving Simulations (SRS) are performed using Python-based Flux Reconstruction solver (PyFR) to investigate shock boundary layer interaction (SWBLI) where an oblique shock impinges on a flat plate boundary layer. The flux reconstruction method is used for its high parallel computational efficiency and a novel positivity-preserving, entropy-based filter is employed for its robust shock capturing. This study focuses on transitional SWBLI regime, which remains poorly understood. Laminar and Turbulent SWBLI have been studied extensively. Simulations are conducted at Mach 2 flow based on classic experiments of Hakkinen in 1950s by employing two distinct setups: the boundary layer-imposed inlet (BLI) and the viscous leading-edge interaction (LEI) configurations. Although numerical SWBLI studies have largely employed BLI setup, the LEI setup has shown better agreement with experimental results for laminar SWBLI. Both setups are used to systematically investigate transitional SWBLI with particular emphasis on the mechanism of shock-induced boundary layer transition and the associated unsteadiness of the separation bubble.

MFMM 2026 Participants

Participants with * next to their name have expressed an interest in joining for dinner. Only if your plans have changed (either way), please email Tom Sykes (thomas.sykes@warwick.ac.uk).

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Larry Godwin	Aston University
Alireza Keyhani-Asl	Aston University
Rebecca Lewis	Aston University
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Andrew McMullan	Aston University
Philip Trevelyan	Aston University
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Barbara Turnbull	University of Nottingham
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